

CBSE Class 10 Mathematics - Answer Key

1. A bag contains 15 cards numbered from 1 to 15. One card is drawn at random from the bag. The probability that the number on the drawn card is a prime number is

Answer: (B)

Solution:

1. Identify the total number of possible outcomes, which is 15.
2. List the prime numbers between 1 and 15: 2, 3, 5, 7, 11, 13.
3. Count the number of favourable outcomes, which is 6.
4. Apply the probability formula: $P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$.
5. Calculate $P(E) = \frac{6}{15} = \frac{2}{5}$.

Derivation:

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$$

$$n(S) = 15$$

$$E = \{2, 3, 5, 7, 11, 13\}$$

$$n(E) = 6$$

$$P(E) = \frac{6}{15} = \frac{2}{5}$$

2. For a given grouped frequency distribution, the assumed mean method formula is $\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$. If the assumed mean $a = 45$, $\sum f_i d_i = 120$, and $\sum f_i = 30$, then the mean $\bar{x}$ is:

Answer: (C)

Solution:

1. Identify the given values: $a = 45$, $\sum f_i d_i = 120$, and $\sum f_i = 30$.
2. Substitute these values into the assumed mean formula: $\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$.
3. Calculate the fraction: $\frac{120}{30} = 4$.
4. Add the result to the assumed mean: $\bar{x} = 45 + 4 = 49$.

Derivation:

$$\bar{x} = 45 + \frac{120}{30}$$

$$\bar{x} = 45 + 4$$

$$\bar{x} = 49$$

3. Assertion (A): The probability of an impossible event is 0. Reason (R): For any event E , the probability $P(E)$ satisfies $0 \leq P(E) \leq 1$.

Answer: (A)

Solution:

1. An impossible event is an event that cannot occur, so the number of favourable outcomes is 0.
2. The probability of an event is defined as the ratio of the number of favourable outcomes to the total number of equally likely outcomes.
3. Thus, $P(\text{impossible event}) = \frac{0}{n} = 0$. So, Assertion (A) is true.
4. The range of probability for any event E is always between 0 and 1 inclusive, i.e., $0 \leq P(E) \leq 1$. So, Reason (R) is true.
5. Since the probability of an impossible event is a specific case of the general rule that $0 \leq P(E) \leq 1$, Reason (R) correctly explains Assertion (A).

Derivation:

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

For an impossible event, favourable outcomes = 0

$$P(E) = \frac{0}{\text{Total}} = 0$$

$0 \leq P(E) \leq 1$ is the fundamental axiom of probability.

4. Assertion (A): For a given grouped frequency distribution, the class mark x_i of a class interval is calculated as the average of its lower limit and upper limit. Reason (R): The mean of grouped data is calculated using the formula $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$, where x_i represents the class mark of the i^{th} interval.

Answer: (A)

Solution:

1. The class mark (mid-point) of a class interval is defined as $\frac{\text{Lower Limit} + \text{Upper Limit}}{2}$. Thus, Assertion (A) is true.
2. The formula for the mean of grouped data using the direct method is $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$, where x_i is the class mark. Thus, Reason (R) is true.
3. The definition of the class mark is the fundamental basis for using it in the mean formula for grouped data. Therefore, the reason correctly explains the assertion.

Derivation:

$$x_i = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

5. A box contains 40 cards numbered from 1 to 40. If a card is drawn at random from the box, find the probability that the number on the card is a multiple of 5.

Answer: 1/5

Solution:

1. Identify the total number of possible outcomes, which is the total count of cards, $n(S) = 40$.

- Identify the favourable outcomes which are multiples of 5 between 1 and 40, i.e., $\{5, 10, 15, 20, 25, 30, 35, 40\}$.
- Count the favourable outcomes, $n(E) = 8$.
- Calculate the probability using $P(E) = \frac{n(E)}{n(S)} = \frac{8}{40} = \frac{1}{5}$.

Derivation:

$$\begin{aligned} n(S) &= 40 \\ E &= \{5, 10, 15, 20, 25, 30, 35, 40\} \\ n(E) &= 8 \\ P(E) &= \frac{n(E)}{n(S)} = \frac{8}{40} = \frac{1}{5} \end{aligned}$$

- For a given frequency distribution, the sum of frequencies $\sum f_i = 50$ and the sum of the product of frequencies and class marks $\sum f_i x_i = 1750$. Find the mean of the distribution.

Answer: 35

Solution:

- Identify the formula for the mean of grouped data, which is $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$.
- Substitute the given values $\sum f_i x_i = 1750$ and $\sum f_i = 50$ into the formula to calculate the mean.

Derivation:

$$\begin{aligned} \bar{x} &= \frac{\sum f_i x_i}{\sum f_i} \\ \bar{x} &= \frac{1750}{50} \\ \bar{x} &= 35 \end{aligned}$$

- A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears: (i) a two-digit number, (ii) a perfect square number.

Answer: The probability of (i) is $\frac{9}{10}$ and (ii) is $\frac{1}{10}$.

Solution:

- Identify the total number of possible outcomes, which is 90.
- For (i), identify the two-digit numbers from 10 to 90. There are $90 - 9 = 81$ such numbers. The probability is $\frac{81}{90} = \frac{9}{10}$.
- For (ii), identify the perfect squares between 1 and 90. These are 1, 4, 9, 16, 25, 36, 49, 64, 81. There are 9 such numbers. The probability is $\frac{9}{90} = \frac{1}{10}$.

Derivation:

- Total \ outcomes \ $n(S) = 90$

2. (i) \ Two-digit \ numbers \ are \ \{10, 11, \dots, 90\}, $n(E_1) = 90 - 9 = 81$
 3. $P(E_1) = \frac{81}{90} = \frac{9}{10}$
 4. (ii) \ Perfect \ squares \ are \ \{1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2\} = \{1, 4, 9, 16, 25, 36, 49, 64, 81\}, $n(E_2) = 9$
 5. $P(E_2) = \frac{9}{90} = \frac{1}{10}$
8. The following table shows the daily pocket allowance of 50 children in a locality. Find the mean daily pocket allowance using the direct method. — Daily allowance (in Rs.) — 100 — 120 — 120 — 140 — 140 — 160 — 160 — 180 — 180 — 200 — — : — — : — : — : — : — : — : — — Number of children — 12 — 14 — 8 — 6 — 10 —

Answer: Rs. 145.20

Solution:

1. Find the class mark (x_i) for each interval using the formula $x_i = \frac{\text{upper limit} + \text{lower limit}}{2}$.
2. Multiply each frequency (f_i) by its corresponding class mark (x_i) to get $f_i x_i$.
3. Calculate the sum of all frequencies $\sum f_i$ and the sum of all products $\sum f_i x_i$.
4. Apply the direct mean formula $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$ to find the final result.

Derivation:

$$\begin{aligned}
 x_1 &= \frac{100 + 120}{2} = 110, x_2 = 130, x_3 = 150, x_4 = 170, x_5 = 190 \\
 \sum f_i &= 12 + 14 + 8 + 6 + 10 = 50 \\
 \sum f_i x_i &= (12 \times 110) + (14 \times 130) + (8 \times 150) + (6 \times 170) + (10 \times 190) \\
 \sum f_i x_i &= 1320 + 1820 + 1200 + 1020 + 1900 = 7260 \\
 \text{Mean} &= \frac{7260}{50} = 145.2
 \end{aligned}$$

9. A box contains a total of 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears: (i) a two-digit number, (ii) a perfect square number, and (iii) a number divisible by 5.

Answer: (i) $\frac{81}{90} = \frac{9}{10}$, (ii) $\frac{9}{90} = \frac{1}{10}$, (iii) $\frac{18}{90} = \frac{1}{5}$

Solution:

1. Identify the total number of outcomes $n(S) = 90$.
2. For (i), identify the two-digit numbers from 1 to 90. These are numbers from 10 to 90. The count is $90 - 10 + 1 = 81$. Calculate $P(E_1) = \frac{81}{90}$.
3. For (ii), identify perfect squares between 1 and 90. These are $1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2$ (1, 4, 9, 16, 25, 36, 49, 64, 81). The count is 9. Calculate $P(E_2) = \frac{9}{90}$.
4. For (iii), identify multiples of 5 between 1 and 90. This is an arithmetic progression 5, 10, ..., 90. Using $a_n = a + (n - 1)d$, $90 = 5 + (n - 1)5$, so $18 = n$. The count is 18. Calculate $P(E_3) = \frac{18}{90}$.

5. Simplify all fractions to their lowest terms.

Derivation:

$$\begin{aligned}
 n(S) &= 90 \\
 (i) \text{ Two-digit numbers: } \{10, 11, \dots, 90\}, n(E_1) &= 81 \\
 \Rightarrow P(E_1) &= \frac{81}{90} = \frac{9}{10} \\
 (ii) \text{ Perfect squares: } \{1, 4, 9, 16, 25, 36, 49, 64, 81\}, n(E_2) &= 9 \\
 \Rightarrow P(E_2) &= \frac{9}{90} = \frac{1}{10} \\
 (iii) \text{ Divisible by 5 : } \{5, 10, \dots, 90\}, n(E_3) &= \frac{90}{5} = 18 \\
 \Rightarrow P(E_3) &= \frac{18}{90} = \frac{1}{5}
 \end{aligned}$$

10. The following table shows the daily pocket money received by 50 students in a locality. If the mean pocket money is Rs.18, find the missing frequencies f_1 and f_2 given that $f_1 + f_2 = 12$. The distribution of pocket money (in Rs.) is: Class Interval: 11 – 13, 13 – 15, 15 – 17, 17 – 19, 19 – 21, 21 – 23, 23 – 25 Frequency: 3, 6, 9, f_1 , f_2 , 5, 4

Answer: $f_1 = 8, f_2 = 4$

Solution:

- Step 1: Calculate the class marks (x_i) for each interval by taking the average of the upper and lower limits.
- Step 2: Construct a table with columns: Class Interval, Frequency (f_i), Class Mark (x_i), and Product ($f_i x_i$).
- Step 3: Calculate the sum of frequencies Σf_i and the sum of products $\Sigma f_i x_i$ in terms of f_1 and f_2 .
- Step 4: Use the formula Mean = $\frac{\Sigma f_i x_i}{\Sigma f_i}$ and the given mean value of 18 to form a linear equation.
- Step 5: Use the given constraint $f_1 + f_2 = 12$ to solve the system of two linear equations in two variables.
- Step 6: Substitute $f_2 = 12 - f_1$ into the mean equation to find f_1 , then calculate f_2 .

Derivation:

$$\begin{aligned}
 x_i &\in \{12, 14, 16, 18, 20, 22, 24\} \\
 \Sigma f_i &= 3 + 6 + 9 + f_1 + f_2 + 5 + 4 = 27 + f_1 + f_2 \\
 \text{Since } \Sigma f_i &= 50, \text{ then } 27 + f_1 + f_2 = 50 \\
 \Rightarrow f_1 + f_2 &= 23 \text{ (Wait, check total sum: } 3+6+9+5+4=27, 50-27=23. \text{ Given } f_1 + f_2 = 12 \text{ is inconsistent with total 50. Let's use total = 39)} \\
 \Sigma f_i x_i &= (3 \times 12) + (6 \times 14) + (9 \times 16) + (f_1 \times 18) + (f_2 \times 20) + (5 \times 22) + (4 \times 24) \\
 \Sigma f_i x_i &= 36 + 84 + 144 + 18f_1 + 20f_2 + 110 + 96 = 470 + 18f_1 + 20f_2 \\
 \frac{470 + 18f_1 + 20f_2}{39} &= 18 \\
 470 + 18f_1 + 20f_2 &= 702 \\
 \Rightarrow 18f_1 + 20f_2 &= 232 \\
 \Rightarrow 9f_1 + 10f_2 &= 116 \\
 \text{Using } f_1 + f_2 &= 12 \\
 \Rightarrow f_2 &= 12 - f_1 \\
 9f_1 + 10(12 - f_1) &= 116 \\
 \Rightarrow 9f_1 + 120 - 10f_1 &= 116 \\
 -f_1 &= -4 \\
 \Rightarrow f_1 &= 4, f_2 = 8
 \end{aligned}$$

11. A box contains a total of 60 cards numbered from 1 to 60. Some cards are marked with red ink, some with blue ink, and some with green ink. The number of red cards is 12 more than the number of green cards, and the number of blue cards is 8. A card is drawn at random from the box.

(i) If the number of green cards is x , find the value of x .

Answer: (B) **Solution:**

1. Let number of green cards be x .
2. Number of red cards = $x + 12$.
3. Number of blue cards = 8.
4. Total cards = $x + (x + 12) + 8 = 60$.
5. $2x + 20 = 60 \Rightarrow 2x = 40 \Rightarrow x = 20$.

Derivation:

$$\begin{aligned}x + (x + 12) + 8 &= 60 \\2x + 20 &= 60 \\2x &= 40 \\x &= 20\end{aligned}$$

(ii) Find the probability that the card drawn is a red card.

Answer: 8/15 **Solution:**

1. Number of green cards $x = 20$.
2. Number of red cards = $20 + 12 = 32$.
3. Total cards = 60.
4. Probability = $32/60 = 8/15$.

Derivation:

$$P(\text{Red}) = \frac{32}{60} = \frac{8}{15}$$

(iii) What is the probability that the card drawn is NOT a blue card?

Answer: (C) **Solution:**

1. $P(\text{Blue}) = 8/60 = 2/15$.
2. $P(\text{Not Blue}) = 1 - P(\text{Blue}) = 1 - 2/15 = 13/15$.

Derivation:

$$P(\text{Not Blue}) = 1 - \frac{8}{60} = 1 - \frac{2}{15} = \frac{13}{15}$$

(iv) Find the probability that the card drawn is either green or blue.

Answer: 7/15 **Solution:**

1. Number of green = 20, number of blue = 8.
2. Number of green or blue = $20 + 8 = 28$.

3. Probability = $28/60 = 7/15$.

Derivation:

$$P(\text{Green} \cup \text{Blue}) = \frac{20 + 8}{60} = \frac{28}{60} = \frac{7}{15}$$

12. A survey was conducted by a group of students as part of their environmental awareness project, in which they collected data regarding the number of plants in 20 houses in a locality. The data is summarized in the following frequency distribution table:
- | Number of plants | 0 – 2 | 2 – 4 | 4 – 6 | 6 – 8 | 8 – 10 | 10 – 12 | 12 – 14 |
|----------------------------|-------|-------|-------|-------|--------|---------|---------|
| Number of houses (f_i) | 1 | 2 | 1 | 5 | k | 2 | 3 |
- The total number of houses is 20.

(i) What is the value of the missing frequency k ?

Answer: (C) Solution:

1. The sum of all frequencies is given as 20.
2. Calculate $1 + 2 + 1 + 5 + k + 2 + 3 = 20$.
3. $14 + k = 20$.
4. $k = 6$.

Derivation:

$$\begin{aligned} 1 + 2 + 1 + 5 + k + 2 + 3 &= 20 \\ 14 + k &= 20 \\ k &= 6 \end{aligned}$$

(ii) Find the class mark (x_i) for the class interval 8 – 10.

Answer: 9 Solution:

1. The class mark is the average of the upper and lower limits.
2. Class mark = $(8 + 10)/2 = 18/2 = 9$.

Derivation:

$$x_i = \frac{8 + 10}{2} = 9$$

(iii) Which method is most suitable to calculate the mean of this grouped data?

Answer: (A) Solution:

1. Since the class intervals are uniform ($h = 2$) and the frequencies/class marks are small, the Direct Method or Assumed Mean Method are both efficient.
2. Given the small numerical values, the Direct Method is straightforward and sufficient.

Derivation:

The values are small, hence the Direct Method $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$ is efficient.

(iv) Calculate the mean number of plants per house.

Answer: 8.1 **Solution:**

1. List class marks (x_i): 1, 3, 5, 7, 9, 11, 13.
2. Calculate $f_i x_i$: $1(1), 2(3), 1(5), 5(7), 6(9), 2(11), 3(13) = 1, 6, 5, 35, 54, 22, 39$.
3. Sum $\sum f_i x_i = 162$.
4. Mean $= 162/20 = 8.1$.

Derivation:

$$\sum f_i x_i = 1 + 6 + 5 + 35 + 54 + 22 + 39 = 162$$

$$\bar{x} = \frac{162}{20} = 8.1$$